

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

2 September 2013

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Circle the question number of the question attempted.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
Total	

This document consists of **23** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 In translation, mRNA that is produced in transcription is used as the template for the synthesis of a polypeptide chain which will be further coiled and folded to form a functional protein. Fig. 1.1 shows a stage during translation.

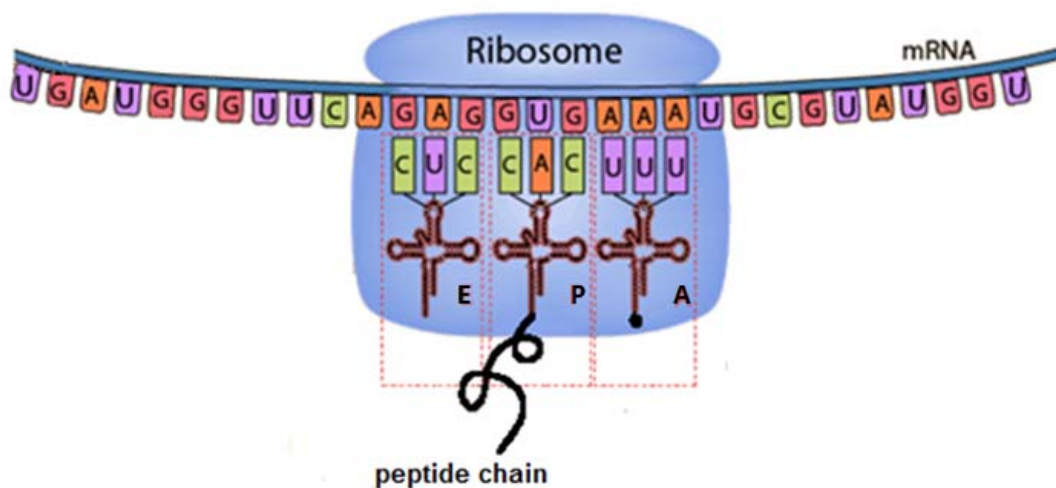


Fig. 1.1

- (a) Name the stage of translation shown in Fig. 1.1. [1]

- (b) State the number of amino acids found on the peptide chain. [1]

- (c) During translation, Enzyme X was used to catalyse the reaction shown in Fig. 1.2.

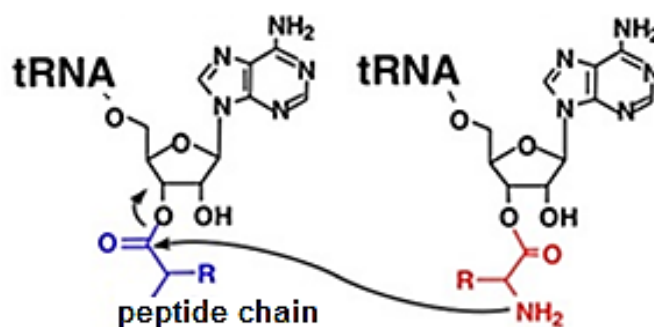


Fig. 1.2

- (i) State the name of Enzyme X. [1]

- (ii) Using the information provided in Fig. 1.1 and Fig. 1.2, describe the process catalysed by Enzyme X. [3]

- (iii) Enzyme X is able to act on all the aminoacyl-tRNA found in the body. Describe the model by which the aminoacyl-tRNA binds to the enzyme. [2]

The rate of reaction of Enzyme X is affected by pH within the cytosol where translation is taking place. Fig. 1.3 was the result of an experiment to investigate how pH affected the rate of translation in normal cells.

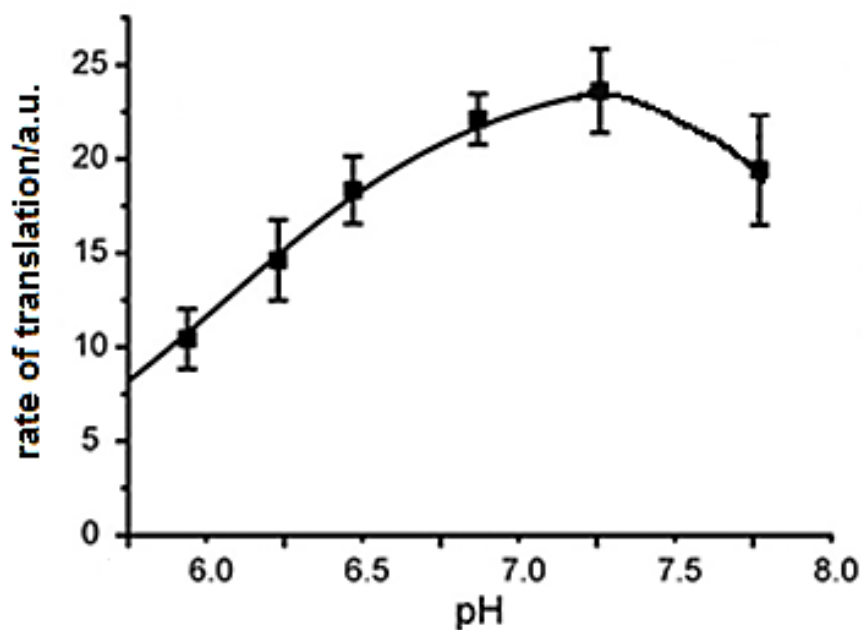


Fig. 1.3

(d) With reference to Fig. 1.3, describe how pH affects the rate of enzyme reaction. [2]

(e) A toxin introduced into normal cells changes the pH of the cells, leading to a reduced rate of translation and hence bringing about cell death.

Suggest what the pH of the cells was after the introduction of the toxin and explain why protein synthesis was affected. [2]

[Total: 12]

- 2 Fig. 2.1 shows a cell that has undergone semi-conservative replication and is in the process of undergoing telophase and cytokinesis.

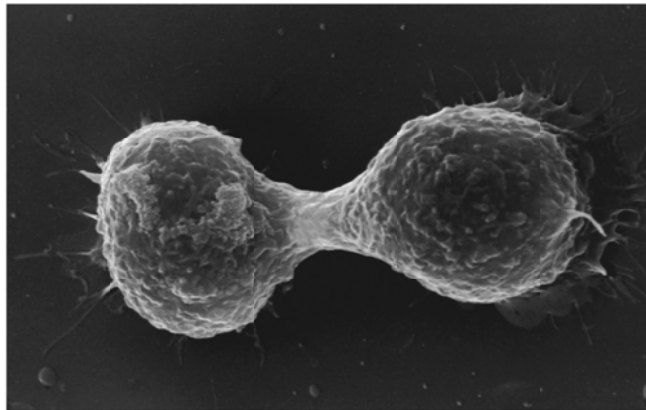


Fig. 2.1

- (a) (i)** State precisely when DNA replication occurs in the cell cycle. [1]

- (ii)** State what is meant by semi-conservation replication. [2]

- (iii)** Describe the roles of enzymes used in the process. [3]

(b) Describe the events that occur during the stage shown in Fig. 2.1. [3]

An enzyme called telomerase is produced by some cells in the human body. Telomerase adds nucleotides to the ends of chromosomes.

Table 2.1 shows the telomerase activity and telomere length of some types of cell after cell division.

Table 2.1

type of cell	number of cell divisions	telomerase activity	telomere length
bone cell	limited	absent	shortened
cancer cell	unlimited	high	maintained
embryonic stem cell	unlimited	high	maintained
epidermis cell	limited	absent	shortened
liver cell	partially limited	low	slightly shortened
neurone	limited	absent	shortened

(c) Using the information in Table 2.1,

(i) State the consequences of high telomerase activity in a cell. [1]

- (ii) Explain the relationship between high levels of telomerase expression and cancer. [2]

[Total: 12]

- 3 During the first phase of the cell cycle, G1, there is a crucial cell-cycle checkpoint that a cell must pass to begin the process that leads to cell division. If changes in the structure of DNA are detected during G1, the cell cycle is halted until DNA is repaired.

p53 gene located on chromosome 17 codes for p53 and is known as a tumour suppressor gene. Mutations of one of the two *p53* gene cause a hereditary disorder known as Li-Fraumeni syndrome.

Fig. 3.1 is a simplified flow chart showing the sequence of events that occurs after a change in the structure of DNA is detected.

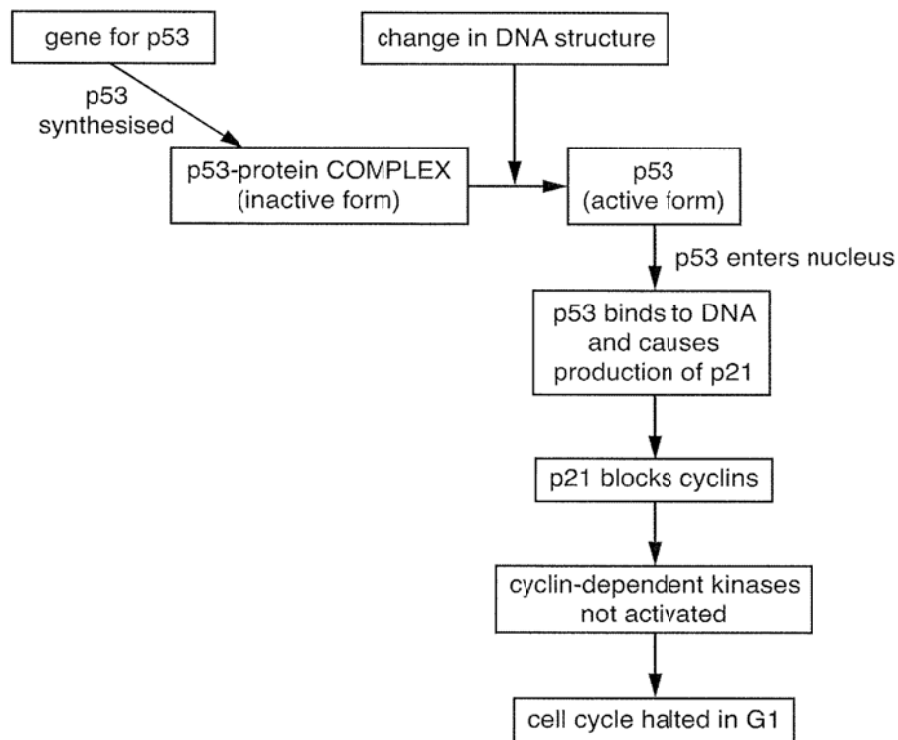


Fig. 3.1

(a) What is a tumour suppressor gene? [1]

In the inactive form, p53 forms a p53-protein complex. In the active form, p53 is no longer attached to the protein.

(b) Suggest why only the active form of p53 can enter the nucleus. [1]

(c) Using the information in Fig. 3.1, explain how mutation of *p53* gene could lead to cells continuing to divide even after a change in the structure of DNA is detected. [3]

The effectiveness of anti-cancer drugs against metastatic colorectal cancer may be determined by growing human tumours in culture. The effectiveness of two drugs, T138067 and vinblastine, on the tumours was assessed. The two drugs were added to the tumours in culture on days 5, 12 and 19. The volumes of the tumours were compared with the volumes of tumours that were not treated with any drugs.

The results are shown in Fig. 3.2.

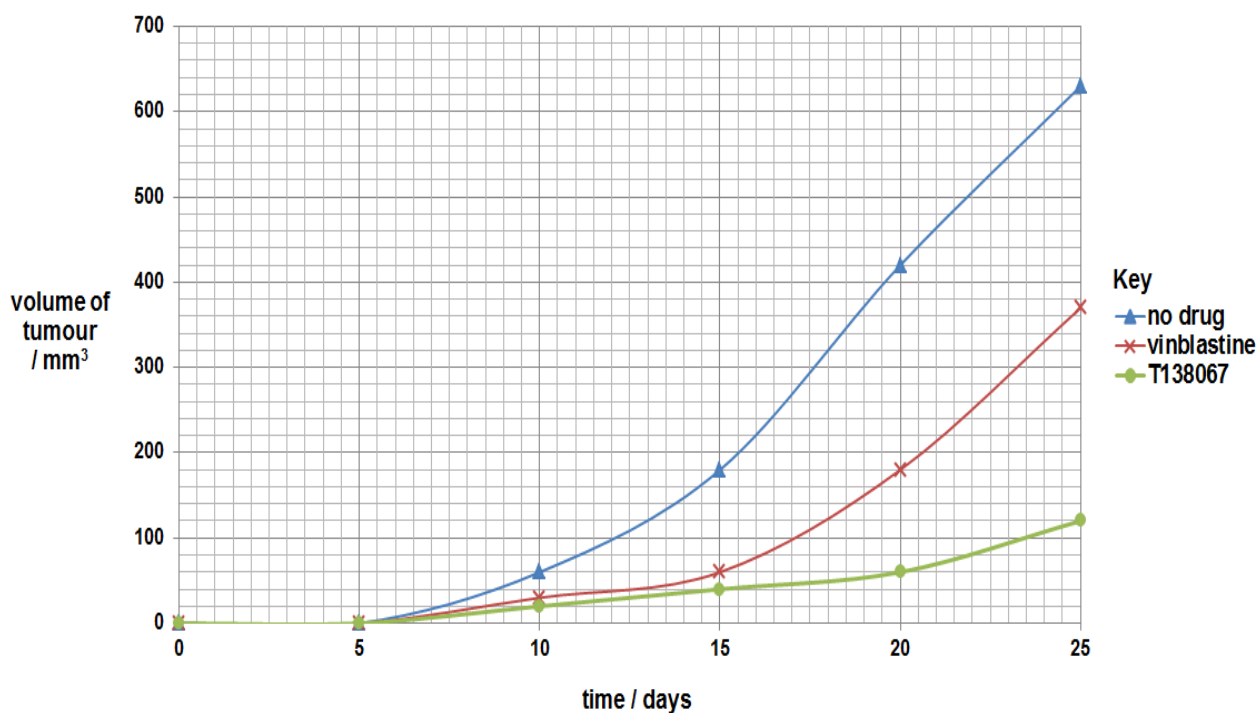


Fig. 3.2

(d) With reference to Fig. 3.2,

(i) conclude the effectiveness of T138067 on tumours. [1]

- (ii) compare the effectiveness of the two drugs when used to treat patients suffering from metastatic colorectal cancer. [2]

Vinblastine disrupts the formation of the spindle apparatus

- (e) Explain how vinblastine acts as an anti-cancer drug. [2]

[Total: 10]

4 Fig. 4.1 shows a Human Immunodeficiency Virus.

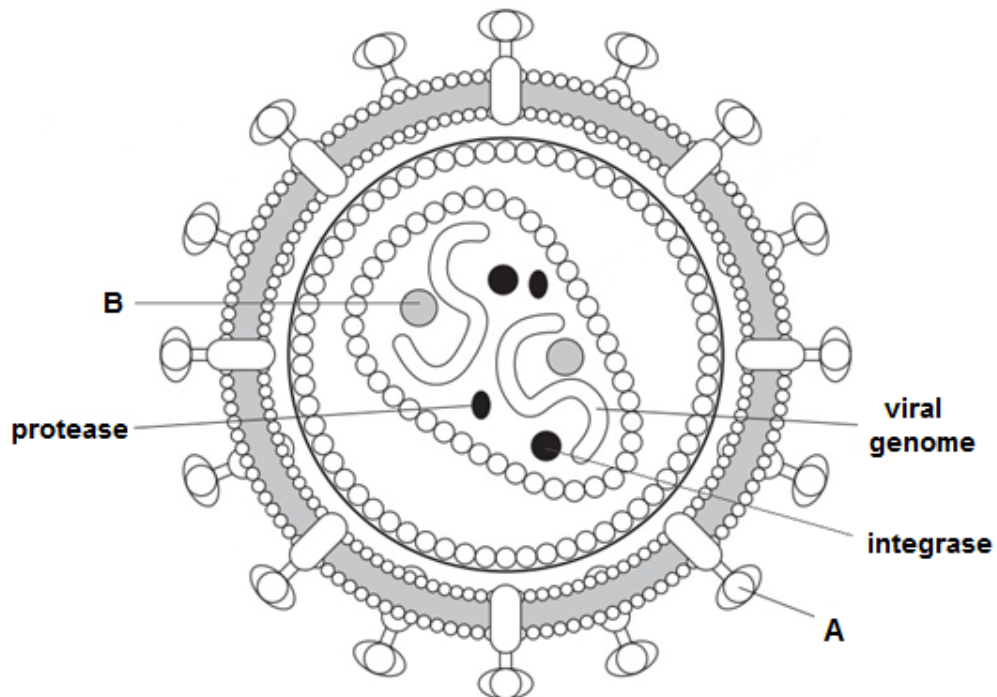


Fig. 4.1

(a) Name the structures A and B in Fig. 4.1. [1]

A _____

B _____

(b) Describe how the viral genome is released into host cell after a conformational change has occurred in gp41. [2]

- (c) Fig. 4.2 shows a generalized graph of the relationship between HIV RNA copies and T lymphocytes counts over the average course of untreated HIV infection.

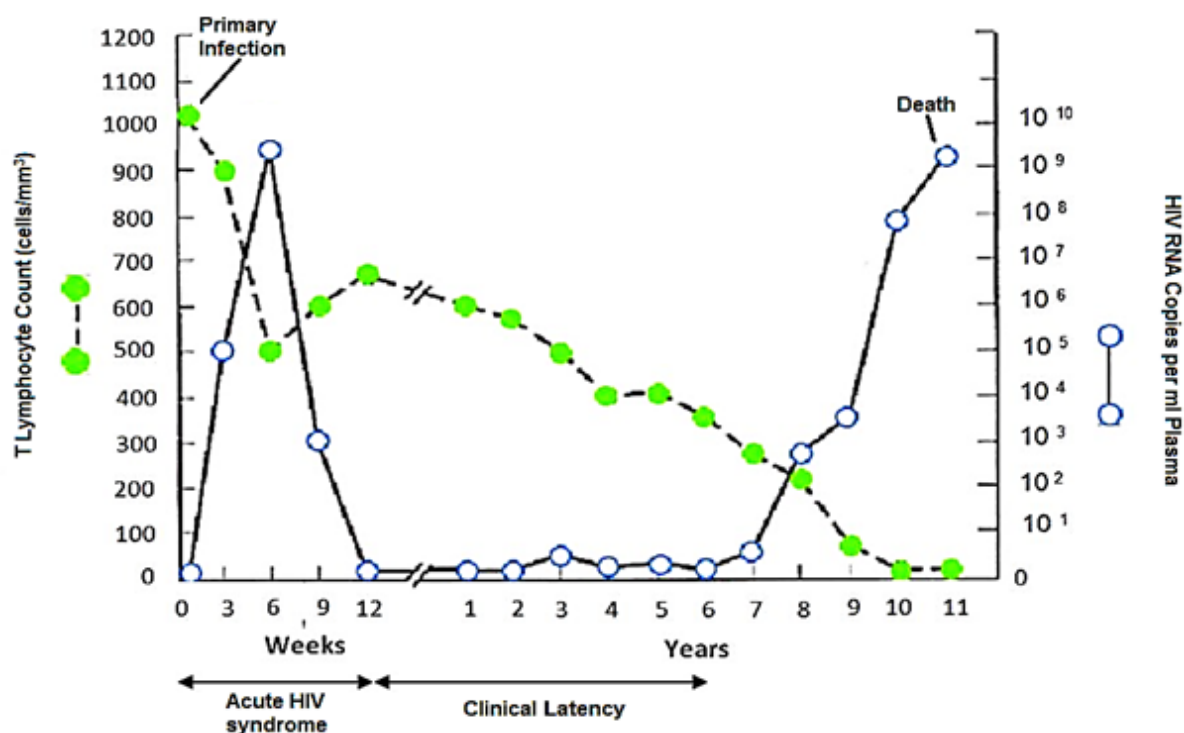


Fig. 4.2

- (i) With reference to Fig. 4.2, describe and explain what happens during the period of clinical latency. [3]

(ii) Explain how HIV infection causes death in the 11th year from primary infection. [2]

(d) The management of AIDS normally includes the use of multiple anti-retroviral drugs that act on different stages of the HIV life cycle in an attempt to control HIV infection. One such drug is alpha interferon which inhibits an enzyme necessary in the maturation and release stage of the HIV life cycle.

Identify the enzyme and explain how HIV infection is controlled. [2]

[Total: 10]

BLANK PAGE

5 A plant breeder carried out three crosses as follows:

- **cross 1** - pure breeding tomato plants with uniformly pigmented red-coloured fruit were crossed with pure breeding tomato plants with dark patches at the base of the orange-coloured fruit
- **cross 2** - the offspring of cross 1 (offspring 1) were crossed with each other
- **cross 3** - offspring 1 were crossed with pure breeding tomato plants with dark patches at the base of the orange-coloured fruit

The number of offspring in each phenotype was counted.
Table 5.1 shows the results of the three crosses.

Table 5.1

cross	parental phenotype		total offspring	number of offspring of each phenotype
1	pure breeding plants with red uniformly pigmented fruit	X pure breeding plants with orange fruit with dark patches	150	150 red uniformly pigmented fruit
2	Offspring 1 plants with red uniformly pigmented fruit	X Offspring 1 plants with red uniformly pigmented fruit	172	24 red fruit with dark patches 98 red uniformly pigmented fruit 46 orange fruit with dark patches 4 orange uniformly pigmented fruit
3	Offspring 1 plants with red uniformly pigmented fruit	X pure breeding plants with orange fruit with dark patches	210	23 red fruit with dark patches 83 red uniformly pigmented fruit 85 orange fruit with dark patches 19 orange uniformly pigmented fruit

(a) Using the symbols **R** for red fruit and **r** for orange fruit, **D** for uniform pigment and **d** for dark patches, draw a genetic diagram to explain the results shown in Cross 3. [4]

(b) Explain the deviation of the observed results from the expected dihybrid ratio. [3]

Observations showed that in tomato plants, there are 3 different types of pigmentation at the base of the fruit. Pigmentation in these fruits is controlled by 2 proteins involved in the same metabolic pathway for production of pigments, Inhibitor *E* and Enzyme *B*. Uniform pigmentation is produced in the presence of two copies of 'e' alleles and one copy of 'B' alleles. Dark patches are produced in the presence of two copies of 'e' alleles. No pigmentation is produced in the presence of at least one copy of 'E' allele.

(c) Suggest a possible mode of inheritance for pigmentation at the base of the fruit. [1]

(d) Explain how the interaction between the two genes *E* and *B* gives rise to pigmentation. [3]

(e) Explain the biochemical basis for this form of epistasis. [2]

[Total: 13]

- 6 Fig. 6.1 shows an electron micrograph of a double membranous organelle, mitochondrion.



Fig. 6.1

- (a) Describe the structure of membrane. [5]

- (b) From your biological knowledge, explain the adaptation of the mitochondrion's double membrane for its role in the production of energy. [2]

Fig. 6.2 shows a stage in respiration that takes place during the oxidation of glucose.

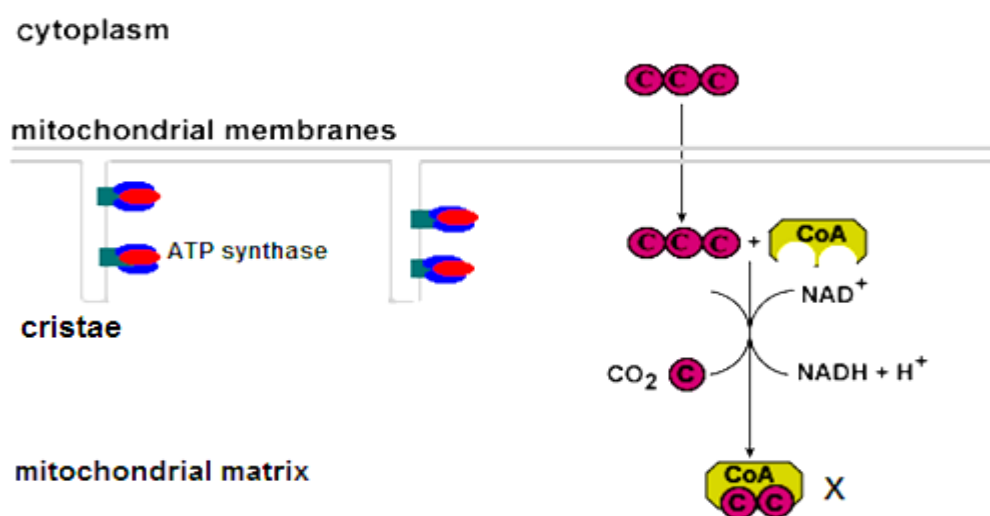


Fig. 6.2

(c) Explain what happens to Molecule X. [3]

(d) List two ways in which photophosphorylation differs from oxidative phosphorylation. [2]

[Total: 12]

- 7 The takahē, *Porphyrio hochstetteri*, is a flightless bird that is restricted to a small area of the South Island of New Zealand. It is one of only two remaining species of large, flightless, herbivorous birds from New Zealand. The other remaining species is found on the North Island of New Zealand. Their flighted ancestor came over from Australia millions of years ago.

The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote mountains of the South Island. The takahē is a grassland specialist and lives in alpine grassland dominated by tussock grasses as shown in Fig. 7.1.



Fig. 7.1

- (a) Explain why it has been possible for flightless birds, such as the takahē, to evolve in New Zealand. [3]

(b) Differences in the DNA sequences that encode cytochrome b in the two species of flightless birds from New Zealand have been measured.

(i) When differences in cytochrome b gene are plotted against time since colonisation, the plot of the line is straight. Describe how this supports the neutral theory of molecular evolution. [2]

(ii) Suggest why the cytochrome b gene is used to measure changes in DNA sequences between the two species of flightless birds. [2]

(c) The takahē is similar in appearance to another flightless bird found on the Desventuradas Islands off the coast of Chile even though molecular studies have shown that they do not share a common ancestor. Explain how biogeography support the evolutionary deductions based on molecular homology. [2]

Another method to determine evolutionary relationships of different animals would be based on anatomical homology.

Fig. 7.2 shows the relationship between some animals which was determined by comparing the arrangement of the bones in their forelimbs, which had a basic pentadactyl structure.

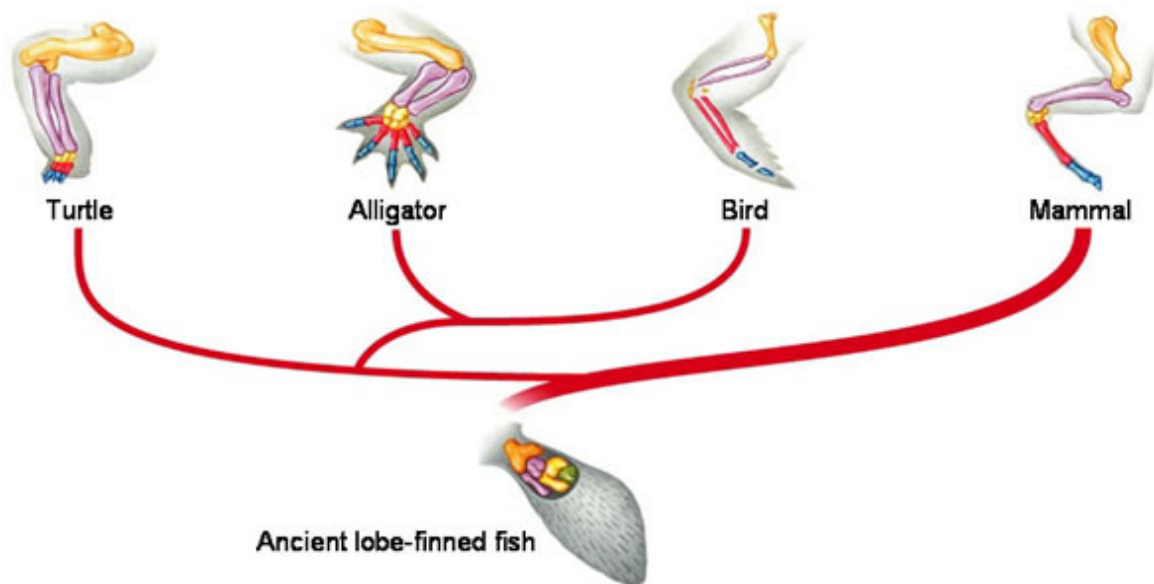


Fig. 7.2

(d) Explain how the anatomical homology in Fig. 7.2 provides evidence to support the theory of evolution. [2]

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Describe the main structural features and function(s) of collagen and glycogen. [8]
(b) Explain how a change in the nucleotide sequence of DNA leads to cystic fibrosis. [8]
(c) Explain how the recessive allele that results in cystic fibrosis may be preserved in a natural population. [4]

[Total: 20]

- 9 (a) Describe the main structural features of the *lac* operon and explain how the presence of lactose in the growth medium affects the *lac* operon. [6]
(b) Describe the role of insulin and glucagon in regulating blood glucose concentration. [8]
(c) Explain how a nerve impulse is passed across a cholinergic synapse. [6]

[Total: 20]